ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET first paper. Section A (30): multiple choice, true or false and matching · Section B (70, seven questions of 10): RL circuits and TRIACs, transformer losses and cells with internal resistance, oscillators and 555 timers, Zener regulators and diode testing, oscilloscope readings, Boolean algebra, and DC generators and motors. Formula sheet at the end.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 The total opposition to alternating current in a circuit that contains resistance and reactance is called …

A resistance. B reactance. C conductance. D impedance.

1.2 The inductive reactance of a coil increases when the frequency …

A increases. B decreases. C becomes zero. D is reversed.

1.3 The RMS value of a sine wave with a peak value of 10 V is …

A 10 V. B 14,14 V. C 7,07 V. D 6,37 V.

1.4 An SCR is switched on by a pulse applied to its …

A anode. B gate. C cathode. D collector.

1.5 Standard ASCII represents each character with … bits.

A 7 B 4 C 16 D 32

1.6 Because of its internal resistance, the terminal voltage of a cell … when it supplies a load.

A rises B stays constant C doubles D drops

1.7 A Karnaugh map is used to …

A measure frequency. B simplify Boolean expressions.
C store data. D amplify signals.

1.8 A 555 timer connected in astable mode produces …

A a steady DC level. B a single pulse.
C a continuous square wave. D a pure sine wave.

1.9 Capacitive reactance is calculated with …

A $X_C = 1 \div (2\pi fC)$.
B $X_C = 2\pi fL$.
C $X_C = R \div Z$.
D $X_C = V \times I$.

1.10 A compound generator has …

A only a series field.
B both a series and a shunt field.
C no field winding.
D only a permanent magnet.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 At resonance in a series RLC circuit, the impedance is equal to the resistance.

2.2 The peak-to-peak voltage of a sine wave is half of its peak voltage.

2.3 A DIAC has three terminals.

2.4 De Morgan's theorems make it possible to build any logic function from NAND gates only.

2.5 The copper losses of a transformer depend on the load current.

2.6 Identical cells connected in parallel give a higher voltage than one cell.

2.7 A crystal oscillator produces a very stable frequency.

2.8 The form factor of a sine wave is about 1,11.

2.9 A DC series motor should be started without any load.

2.10 An oscilloscope can display both AC and DC voltages.

(10)

[10]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Instrument that displays the shape of a waveform	A	form factor
3.2	Opposition of a capacitor to alternating current	B	ASCII
3.3	Four-layer device that is switched on through its gate	C	oscilloscope
3.4	Code that represents letters and symbols as binary numbers	D	voltage regulator
3.5	Circuit that produces a continuous waveform without an input signal	E	commutative law
3.6	Keeps the output voltage constant when the load changes	F	SCR

3.7	A + B = B + A	G	frequency
3.8	Generator whose field current comes from its own armature	H	oscillator
3.9	Ratio of the RMS value to the average value	I	capacitive reactance
3.10	Number of complete cycles in one second	J	self-excited generator

(10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 A coil with an inductance of 0,2 H is connected in series with a 40 Ω resistor across a 230 V, 50 Hz supply. Calculate the:

4.1.1 inductive reactance (2)

4.1.2 impedance (3)

4.1.3 current (2)

4.2 List THREE applications of a TRIAC. (3)

[10]

QUESTION 5

5.1 Name TWO types of losses in a transformer, and explain how EACH is reduced. (4)

5.2 Six cells, each with an emf of 1,5 V and an internal resistance of 0,1 Ω, are connected in series across a 5,4 Ω load. Calculate the total emf, the total internal resistance, the current and the terminal voltage. (6)

[10]

QUESTION 6

6.1 Name the THREE parts (requirements) that every oscillator needs in order to produce a continuous output. (3)

6.2 A 555 timer in astable mode has R_1 = 1 kΩ, R_2 = 10 kΩ and C = 10 µF. Calculate the output frequency. (4)

6.3 Give THREE uses of monostable multivibrators. (3)

[10]

QUESTION 7

7.1 A 5,1 V Zener diode is used to regulate a 12 V supply. The load draws 20 mA and the Zener current must be 5 mA. Calculate the value of the series resistor. (3)

7.2 Explain how the Zener diode keeps the load voltage constant when the supply voltage rises. (3)

7.3 Explain how to test a diode with a digital multimeter, and how to tell whether it is good or faulty. (4)

[10]

QUESTION 8

An oscilloscope is set to 2 V/div and 0,5 ms/div. The sine wave on the screen rises 3,5 divisions above the centre line, and one complete cycle is 4 divisions wide. Calculate the:

8.1 peak voltage (2)

8.2 peak-to-peak voltage (2)

8.3 RMS voltage (2)

8.4 period (2)

8.5 frequency (2)

[10]

QUESTION 9

9.1 State De Morgan's TWO theorems. (4)

9.2 Simplify the Boolean expression $F = A \cdot B + A \cdot \overline{B}$. (2)

9.3 Draw the logic diagram for $F = (A + B) \cdot \overline{C}$. (4)

[10]

QUESTION 10

10.1 Explain the difference between a separately excited and a self-excited DC generator. (4)

10.2 A DC shunt motor is connected to 220 V. Its armature current is 20 A and its armature resistance is 0,5 Ω. Calculate the back emf. (3)

10.3 Explain why a DC series motor must always be started with a load connected. (3)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Inductive reactance: $X_L = 2\pi fL$	Capacitive reactance: $X_C = 1 \div (2\pi fC)$
Impedance: $Z = \sqrt{(R^2 + X^2)}$	Ohm's law (AC): $I = V \div Z$
$V_{rms} = 0{,}707 \times V_p$; $V_{av} = 0{,}637 \times V_p$	$V_{p\text{-}p} = 2 \times V_p$; form factor = $V_{rms} \div V_{av}$
Frequency: $f = 1 \div T$	555 astable: $f = 1{,}44 \div ((R_1 + 2R_2) \times C)$
Cells in series: $E_T = nE$; $r_T = nr$	$I = E_T \div (R + r_T)$; $V = I \times R$
Zener series resistor: $R = (V_{in} - V_Z) \div (I_Z + I_L)$	DC motor: $E_b = V - I_aR_a$; generator: $E = V + I_aR_a$